

**SVKM's NMIMS (Deemed-to-be University)**  
**School of Technology Management & Engineering (STME), Indore**  
**TERM END EXAMINATION — MARCH 2026**  
**Dynamic Systems Modeling (702MH0C025)**

|                              |              |                   |                    |
|------------------------------|--------------|-------------------|--------------------|
| Program: B.Tech Mechatronics | Semester: IV | Duration: 3 Hours | Maximum Marks: 100 |
|------------------------------|--------------|-------------------|--------------------|

**Instructions to Students:**

1. **Q1 is compulsory** (20 marks) — attempt all four sub-parts.
2. From Q2 to Q7, **attempt any FOUR** questions ( $4 \times 20 = 80$  marks).
3. Draw neat **Free Body Diagrams** wherever applicable.
4. No programmable calculators. Use  $g = 9.81 \text{ m/s}^2$  unless stated otherwise.
5. All numerical answers must include **proper units**.
6. Q4 & Q5 contain both a **theory (10M)** and a **numerical (10M)** part.
7. Q6 & Q7 contain an **FBD analysis (10M)** and a **difficult numerical (10M)**.

**Marking Scheme & Syllabus Coverage**

| Q. No.       | Description                                                   | Unit(s)     | CO              | Bloom's | Marks     | Type       | Difficulty         |
|--------------|---------------------------------------------------------------|-------------|-----------------|---------|-----------|------------|--------------------|
| <b>Q1</b>    | Compulsory — Attempt ALL 4 sub-parts                          | <b>1–6</b>  | <b>CO1–CO5</b>  | L2–L3   | <b>20</b> | Mixed      | <b>Easy–Medium</b> |
| <b>Q1(a)</b> | Theory: Definitions / Core Concepts                           | <b>1, 3</b> | <b>CO1</b>      | L2      | <b>5</b>  | Theory     | <b>Easy</b>        |
| <b>Q1(b)</b> | Numerical: Particle Kinematics                                | <b>1</b>    | <b>CO1</b>      | L3      | <b>5</b>  | Numerical  | <b>Easy</b>        |
| <b>Q1(c)</b> | Numerical: Constrained Motion / Pulleys                       | <b>4</b>    | <b>CO3</b>      | L3      | <b>5</b>  | Numerical  | <b>Medium</b>      |
| <b>Q1(d)</b> | Numerical: Energy / Rigid Body                                | <b>5, 6</b> | <b>CO5</b>      | L3      | <b>5</b>  | Numerical  | <b>Medium</b>      |
| ■■■          | ■■■                                                           | ■■■         | ■■■             | ■■■     | ■■■       | ■■■        | ■■■                |
| <b>Q2</b>    | Pure Numerical: Particle Kinetics (4 parts $\times$ 5M)       | <b>4</b>    | <b>CO3, CO4</b> | L3      | <b>20</b> | Numerical  | <b>Medium</b>      |
| <b>Q3</b>    | Pure Numerical: Rigid Body & Vibrations (4 parts $\times$ 5M) | <b>5</b>    | <b>CO4, CO5</b> | L3      | <b>20</b> | Numerical  | <b>Medium</b>      |
| <b>Q4</b>    | Theory (10M) + Numerical (10M): Kinematics / Rotating Frames  | <b>3</b>    | <b>CO1, CO3</b> | L4      | <b>20</b> | Theory+Num | <b>Medium</b>      |
| <b>Q5</b>    | Theory (10M) + Numerical (10M): Energy Methods / Lagrangian   | <b>2, 6</b> | <b>CO4, CO5</b> | L4      | <b>20</b> | Theory+Num | <b>Hard</b>        |
| <b>Q6</b>    | FBD (10M) + Difficult Numerical (10M): Applied Dynamics       | <b>4, 5</b> | <b>CO2, CO3</b> | L5      | <b>20</b> | FBD+Num    | <b>Hard</b>        |
| <b>Q7</b>    | FBD (10M) + Difficult Numerical (10M): Advanced Systems       | <b>5, 6</b> | <b>CO2, CO5</b> | L5      | <b>20</b> | FBD+Num    | <b>Hard</b>        |

**Marks Distribution**

| Section            | Questions                                    | Marks per Q   | Total Marks      |
|--------------------|----------------------------------------------|---------------|------------------|
| Q1 — Compulsory    | Q1(a) + Q1(b) + Q1(c) + Q1(d)                | 5 marks each  | 20 marks         |
| Q2–Q7 — Elective   | Attempt any FOUR from Q2, Q3, Q4, Q5, Q6, Q7 | 20 marks each | 80 marks         |
| <b>Grand Total</b> |                                              |               | <b>100 marks</b> |

**Course Outcomes (COs) Assessed**

| CO         | Description                                                 | Questions          |
|------------|-------------------------------------------------------------|--------------------|
| <b>CO1</b> | Understand & apply kinematics of particles and rigid bodies | Q1(a), Q1(b), Q4   |
| <b>CO2</b> | Analyse dynamic systems using Free Body Diagrams            | Q6(a), Q7(a) [FBD] |
| <b>CO3</b> | Calculate forces acting on dynamic systems                  | Q1(c), Q2, Q4, Q6  |
| <b>CO4</b> | Apply constitutive laws to dynamic systems                  | Q1(d), Q2, Q3, Q5  |

|     |                                                      |                   |
|-----|------------------------------------------------------|-------------------|
| CO5 | Calculate energy, work, and power in dynamic systems | Q1(d), Q3, Q5, Q7 |
|-----|------------------------------------------------------|-------------------|

### Syllabus Units Covered

| Unit   | Title                                 | Relevant Questions                                           |
|--------|---------------------------------------|--------------------------------------------------------------|
| Unit 1 | Kinematics of Particles               | Q1(b) — particle kinematics                                  |
| Unit 2 | Kinetics of Particles — Newton's Laws | Q5 — Newton's laws, work-energy                              |
| Unit 3 | Kinematics of Rigid Bodies            | Q1(a), Q4 — kinematics of rigid bodies, rotating frames      |
| Unit 4 | Kinetics of Rigid Bodies              | Q1(c), Q2, Q6 — kinetics, constrained motion, FBD            |
| Unit 5 | Energy & Momentum Methods             | Q1(d), Q3, Q5, Q6, Q7 — energy/momentum, vibrations, rolling |
| Unit 6 | Vibrations & Oscillations             | Q1(d), Q3, Q5, Q7 — vibrations, Lagrangian methods           |

### Preparation Tips

- **Q1 (compulsory):** Revise all definitions and short numerical from all units. Easy marks!
- **Q2 (Particle Kinetics):** Pulleys, inclined planes, circular motion. Practice  $F = ma$  applications.
- **Q3 (Rigid Body & Vibrations):** Rolling objects, pendulums, spring-mass-damper — know formulas cold.
- **Q4 (Kinematics/Rotating Frames):** Derive velocity and acceleration equations. Know Coriolis.
- **Q5 (Energy/Lagrangian):** Practise Lagrange formulation and work-energy theorem problems.
- **Q6 & Q7 (FBD + Difficult Numerical):** Always draw FBD first — it's worth 10 marks on its own!
- **Recommended strategy:** Attempt Q1 → then pick 4 from Q2–Q7 based on your strongest units.

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This template is provided for preparation purposes only. Exact question numbers and specific numerical values will differ in the actual examination paper. Good luck! — Dr. Sunny Nanade, Course Instructor